
A Study of Resource Constraints and Open-Source LLMs for Data Annotation

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Abstract

In AI, data is the central lifeline. The quality, diversity, and volume of data determine how intelligent and reliable a model can become. Therefore, labeling the data correctly is very crucial for AI. But labeling or annotating data is not an easy task. It takes time, money, effort, and correctness. But in 2025, where AI especially LLM has become an important part of our life, whether we can use LLM for data annotation task is the main focus of our research. In our research, we have selected eight types of NLP tasks with corresponding human labeled dataset and tested them using 10 LLM models. Our main goal is to find out whether LLMs are able to correctly label the data. We have found that different LLMs perform differently in different tasks. In tasks like sentiment analysis, their accuracies are quite remarkable. Also, they performed poorly on other tasks like Relation Extraction. We believe that our research can contribute to the correctness and efficiency of the data annotation process.

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Chapter 1

Introduction

Annotating data is a high resource consuming task. It takes a lot of time, man power & money to annotate a large amount of data, yet still after the completion of annotation there resides possibility of error because human aren't perfect. Even without the aspect of human error even if annotation is done by expert professionals there exists an issue of perspective for which different person might consider a data to fall under a different label. So even if we spend time & resources unless we train the annotators to follow same standard there lies possibility of error in annotation. Standing on 2025, we have many effective LLMs like DeepSeek, Llama, gemma etc. and we wonder if we could use these modern LLMs to annotate our data reliably. If we could reliably use these LLMs for annotation it would be very efficient and helpful for research work. But can LLMs do annotation properly? In order to find answer to this question and to find out which model performs best we will use around ten open source LLMs and compare their results with human annotation, lastly we will compare based on task which LLM does best on which task. Our contribution through this work will be to help in understanding the pros and cons of human annotation and LLM annotation. Which LLM should be used, which to not. Also a comparison of annotating tasks between the LLMs will help people choose the right LLM based on their annotation task.

1.1 Project Overview

The objective of this project is to revolutionize the data annotation pipeline by leveraging the power of LLMs such as Qwen3, Gemini, LLaMA and others to automate tasks that have historically been accomplished through manual or human-driven labor. In this work, we propose a framework that could reduce the reliance on humans for annotation. Fundamentally, the system leverages pre-trained LLMs to label text data without any human intervention, based on specially designed measurement metrics and tuning pipelines. For validating the model outputs, we curate data that is fully annotated by humans. This data serves as both a benchmark and a knowledge base for the LLM-based annotation. We target various NLP tasks that require annotation, such as sentiment analysis, natural

language Inference,intent detection, Relation Extraction, emotion classification etc. The llm annotators are evaluated with respect to human annotations in terms of both quality and efficiency (labeling time and accuracy). In the end, by automating annotation without any loss in quality, and freeing people to work on higher-level tasks, we hope it becomes easier and cheaper to create high-quality NLP models.

1.2 Motivation

Limitation of proper natural language dataset is one of the biggest limitation of NLP(natural language processing) models currently. Despite the improvement in machine learning language classification tasks such as name entity recognition(NER), sentiment analysis etc, tasks rely on limited expensive human annotation. This dependence on properly annotated data limits scalability and often makes it difficult to make high-quality research. We plan to contribute to this field by helping in understanding the capabilities of LLMs for annotation through our project, which will allow users to evaluate how different open-source models perform on various NLP tasks. Our paper will test around ten open-source LLMs across approximately eight different tasks. This will provide valuable insights for developers and those looking to integrate LLMs in a resource-efficient manner. This is particularly beneficial for low-resource projects.

1.3 Objectives

The aim of this study is the development of an automated annotation framework, which uses large language models under seven to eight billion parameters, such as Deepseek, Llama, Gamma, Mistral,qwen3,phi-4-mini, Falcon3,pythia, zephyr, XiaomiMiMo. This framework aims to minimize dependence on manual human annotation in natural NLP tasks. The proposed system, therefore, seeks to overcome the limitations of traditional annotation workflows, which are typically time-consuming and costly, and error-prone This can be due to inconsistencies among different annotators. Specific aims for this study are as follows:

1. We design an annotation pipeline that identifies these LLM models' labeling of all the textual data from different NLP tasks such as Sentiment analysis, Hate speech detection, Intent Detection, Paraphrase Detection, Emotion Classification, Natural Language Inference, Relation Extraction and News classification.
2. To build a high-quality benchmark dataset which is fully annotated by human by experts. The data will be used as comparing reference and an evaluation standard. In this setting, human reference annotations would be compared with automatic annotations produced by LLMs.
3. We are comparing the predicted results from the LLM models with the human-annotated datasets using accuracy and F1-score.

4. To improve annotation accuracy by prompt through integrating prompt tuning strategies.

1.4 Methodology

Our project is to build a pipeline that can generate annotations using various LLM models based on different NLP tasks such as Named Entity Recognition (NER), Sentiment Detection, Hate Speech Detection, and more — a total of eight assigned tasks. The LLM models will label the data according to the specific tasks. Therefore, the first step we need to follow:

1. Dataset Preparation: We selected eight tasks Sentiment analysis, Hate speech detection, Intent Detection, Paraphrase Detection, Emotion Classification, Natural Language Inference, Relation Extraction and News classification.

For each task, related datasets are collected, and due to GPU limitations, we will randomly select 1000 sentences per task in some task less than that for small test data size. Each dataset contains fully human-annotated labeled data.

2. LLM Annotation: We will use 10 open-source LLMs (each under 7–8 billion parameters), such as LLaMA, Gamma, Qwen3, Phi4-Mini, DeepSeek, etc. For each task, the LLMs will be prompted according to the task description using zero-shot and few-shot prompting.

Each prompt will contain a short instruction like given a sentence, provide a one-line description of its meaning and classify it into the appropriate label class. For example, for the Natural Language Inference (NLI) task: “Given two short texts, determine their relationship: entailment, contradiction, or neutral.” In a zero-shot prompt, we provide the sentence pair and ask the model to output a one-line explanation and the predicted label (entailment, contradiction, or neutral). The LLM’s outputs for all 1000 sentences per task are collected and stored separately.

3. Evaluation and Benchmark Analysis: For each task, the LLM’s short label response (one-word classification) is compared with the human-annotated labels in the dataset. Based on this comparison:

A confusion matrix is created for each model–task pair. Accuracy and F1-score are calculated. After that a benchmark table is generated showing each model’s performance across all tasks. This benchmark will indicate which LLM performs best for each NLP task based on the evaluation metrics and prompt style.

1.5 Project Outcome

This project aims to develop an automated annotation framework that leverages large language models (LLMs) to perform high-quality annotation across multiple NLP tasks, reducing human effort while maintaining accuracy and efficiency. This project will use large language models (LLMs) like Gamma, LLaMA, Phi-4 mini, and other specially

selected models in its automated annotation framework to perform fundamental NLP tasks such as hate speech detection, sentiment analysis and six others. We present a fully human-annotated superior dataset to evaluate LLMs' performance and show that LLMs can achieve human-level accuracy. Moreover, by timely optimization and uncertainty-based task delegation, they can very quickly bring down the time and cost of annotation. The framework, which harnesses some of the strongest open-source models, is about a modular, scalable, and generalizable pipeline for the efficient use of open-source LLM models in data annotation.

Chapter 2

Background

2.1 Preliminaries

So this project is evaluating large language models based on 7 to 8 billion parameters. We selected the models based on their popularity, download count, and performance. Using these models, we will perform annotation on eight types of tasks and create a benchmark. The best three things that we focus on is :

1. Diversity of LLMs based on tasks for each model.
2. Show the capabilities of LLMs how well they perform in which task compared to human annotation
3. Making a benchmark of those models showing their capabilities and backlog or bindings
4. Showing can LLM fully or partially replace of annotation task.

2.1.1 Tasks And Datasets

In this experimental work, we perform ten types of LLM models to labelize all the tasks. So we sample ten general-purpose labeling text classification tasks where they will show how LLMs performed based on each task. Which model performs better on which task labeling. Over all based on all the tasks that performed better. And showing that this type of LLMs is how efficient doing this type of annotation work using low-label LLM models. Here the task was to identify their label was Sentiment analysis, Hate speech detection, Intent Detection, Paraphrase Detection, Emotion Classification, Natural Language Inference, Relation Extraction and News classification. From each task we chose 1000 sentence randomly to identify their capability.

1. Sentiment Analysis : Our first task was sentiment analysis. In one line we can explain that a plain text needs to determine whether the text conveying vibe is positive or negative. That's a hard task for LLMs sometimes to identify these 2 labels. So for this sentiment analysis, we have chosen IMDb dataset[1]. This IMDb dataset contains a large movie review dataset. That dataset for binary sentiment classification contains substantially more data. It provides a set of 25,000 highly polar movie reviews for training, and 25,000 for

testing. There is additional unlabeled like 50000. So, from this dataset, we have taken the testing dataset for all the experiment .

2. Hate Speech Detection: Basically, hate speech detection has another name is Offensive Language Identification Dataset (OLID). Here sentence is offensive or harmful or harassment etc like that type. So for this type data set we chose this OLID dataset[2]. This dataset contains 14,100 annotated tweets from Twitter, annotated with three subcategories via crowdsourcing which is also a human-annotated dataset. This dataset has 2 labels one is offensive another one is non-offensive. From here we have chosen 1000 samples size . Where 500 are offensive and 500 non-offensive sentence.

3. Intent detection : Intent detection where it's say type of intent of a speech. to identify this task we chosen snips built in intents (SBIN) which was initially build for to compare different voice assistants and released as a public dataset[3]. Here the dataset contains 328 utterances over 10 intent classes which are 'ComparePlaces', 'RequestRide', 'GetWeather', 'SearchPlace', 'GetPlaceDetails', 'ShareCurrentLocation', 'GetTrafficInformation', 'BookRestaurant', 'GetDirections', 'ShareETA'. This dataset is typically Originally prepared by snips.ai. And the annotation process by the business team at Snips.

4. Paraphrase Detection : Paraphrase means there is some sort of sentence which should be sequentially and related to other sentences. All the sentence meanings should be related. For this task, we chose paws dataset[4]. Here it contains 2 different sentences. If these two sentences are related then label will be paraphrased or not paraphrased. This dataset contains 108,463 human-labeled and 656k noisily labeled pairs that feature the importance of modeling structure, context, and word order information for the problem of paraphrase identification. From there we have 49401 training , 8000 validation 8000 test. So from here we chose the test dataset with 1000 samples. The ratio was fifty-fifty.

5. Emotion Classification : What we understand is that emotion means sad, joy, anger etc. For this label of task the dataset used is GOEmotion[5]. This dataset contains there label of emotion like joy, anger sadness and collected it comments extracted from Reddit, with human annotations. Here Size of training dataset: 43,410 , size of test dataset: 5,427, size of validation dataset: 5,426. Here we use the testing dataset.

6. News Classification: News is a very vital thing in our life. It provides joy, anger, sadness. Beside that it gives us national , international news. Also sports news. For this task we chose ag news[6]. This dataset created collection more than 1 million news articles. Gathered more than 2000 news. Here from 7600 testing data we used 1000 dataset. These dataset have world, sports, scitech and business label dataset.

7. Natural language inference : It is also known as recognizing textual entailment (RTE). We used SNLI corpus[7] . Which is a collection of 570k human-written English sentence pairs manually labeled for balanced classification and testing dataset size 10000 with the labels entailment, contradiction, and neutral, supporting the task of natural language inference (NLI). We used each of label from the dataset for this task.

8. Relation Extraction : In this task we do to find out llms where relation between to entities. For this task we used sem eval 2010 task 8 dataset[8]. So, this dataset contain 10717k

.Here it contains this lable Cause-effect,Component-whole,Content-container,Entity destination,Entity origin,Instrument-agency,Member-collection,Message-topic,Product-producer and Others.

2.1.2 Model Selection

We selected ten widely adopted open-source Natural Language Generation (NLG) models representing diverse architectural designs, parameter scales, and instruction-tuning strategies. All models are publicly available and commonly used in contemporary NLP research.

1. DeepSeek-LLM-7B-Base : DeepSeek-LLM-7B-Base is a decoder-only transformer with 7 billion parameters created by DeepSeek AI. It is trained on large corpuses of multilingual text and optimized to perform reasoning and code-generating tasks. There is no peer-review publication for this checkpoint, nor will one ever be published, so we cite the documentation of its official release. [9].

2. Llama-3.1-8B-Instruct : Llama-3.1-8B-Instruct is part of the LLaMA 3 family introduced by Meta AI. It is an 8B parameter instruction-tuned model trained with improved alignment and data curation techniques. The foundational architecture and training methodology are described in the LLaMA 3 technical report [10].

3. Mistral-7B-Instruct-v0.3 : Mistral-7B-Instruct is an instruction-tuned version of the Mistral 7B base model developed by Mistral AI. The model introduces grouped-query attention (GQA) and sliding window attention mechanisms to improve efficiency. The architecture is described in [11].

4. Gemma-7B-IT : Gemma-7B-IT is a 7B parameter instruction-tuned model released by Google. It is created for easy deployment and experimentation with research. There is no officially published research; however, we cite the official documentation for the model. [12].

5. Phi-4-Mini-Instruct : Phi-4-Mini-Instruct is an instruction-tuned language model developed by Microsoft. The Phi series focuses on creating realistic text using high-quality training data that has been curated, along with the use of augmentation to produce synthetic data via computer algorithms. [13].

6. Qwen3-8B : Qwen3-8B is an 8B parameter transformer model developed by Alibaba. It is trained on multilingual corpora and optimized for instruction following and reasoning tasks [14].

7. Falcon3-7B-Base : Falcon3-7B-Base is a 7B parameter foundation model developed by the Technology Innovation Institute. It extends the Falcon architecture family designed for large-scale pretraining efficiency [15].

8. Pythia-6.9B : Pythia-6.9B is part of the Pythia model suite developed by EleutherAI. The suite focuses on interpretability and scaling analysis across model sizes. The methodology is described in [16].

9. Zephyr-7B-Beta : Zephyr-7B-Beta is an instruction-tuned model released by Hug-

gingFaceH4, built upon the Mistral-7B base architecture. It is fine-tuned using preference optimization and curated instruction datasets [17].

10. MiMo-7B-Base : MiMo-7B-Base is a 7B parameter transformer introduced by Xiaomi. The MiMo framework aims to improve reasoning performance from pretraining to post-training stages. The technical details are provided in [18].

2.1.3 Evaluation Approach

To evaluate the annotation quality of the selected language models in comparison to human annotations, we employ standard classification evaluation metrics: accuracy, precision, recall, and F1 score. Accuracy provides an overall measure of correctness by calculating the ratio of correctly predicted labels to total predictions. Precision focuses on the quality of positive predictions, indicating how many of the predicted labels were actually correct, while recall measures the model’s ability to identify all relevant instances, reflecting how many actual positive cases were correctly detected. The F1 score, as the harmonic mean of precision and recall, offers a balanced metric that is especially valuable when dealing with imbalanced datasets. Together, these metrics offer a comprehensive evaluation framework for assessing how closely model-generated annotations align with human annotations across various NLP tasks.

2.2 Literature Review

Manual data labelling is among the most important tasks in supervised machine learning pipelines - NLP included. Traditional annotation methods rely entirely on human annotators, and often lead to subjectiveness, inconsistency as well as huge time cost. Chiang and Lee [19] point out that lack of proper guidelines in annotation will lead to a poor quality of final annotation in crowd-sourcing projects. This poor quality also affects how easily interpretable NLP models might be evaluated. To cope with the overhead of manual annotation, pre-annotation approaches have been widely adopted by researchers. Indeed, these methods are facilitating human annotations but keeping them constant (with less effort by the human annotators). Lingren et al. [20] showed that dictionary-based pre-annotation in clinical texts greatly accelerates annotation process and there is no annotation bias, suggesting that a practical benefit in semi-automated workflows. With the emergence of large language models (LLMs), including ChatGPT, LLaMA and Claude or Wuendrickx et al., 2019 Model-based annotation has gained traction. Nasution and Onan [21] investigated LLM-generated of annotation in low-resource high-cost languages and found that the LLMs rival humans on simple tasks but are worse than humans on complex, scenario-dependent ones. They also emphasize the scalability-annotation accuracy trade-off. Gao et al. [22] introduced LLM-based annotation pipelines as they develop while few-shot prompting with filters to improve the label quality. Their results indicate that joint work-loss is best suited in the terms of achieving a rate that is minimally necessary and reaches

its default maximum when providing uncertainty-sensitive filtering (like entropy or model agreement) can facilitate improved LLM annotation quality. Other research like Gilardi et al. [23] has shown the performance of ChatGPT based on political background. They have followed a usable technique and used a text classification method to show that the GPT can perform like human. They also showed that the clear prompt format is vital for annotation agreement. Zhou et al. [24] introduced a model named LIMA, which is fine tuned for alignment with human judgment. In their research it is shown that instruction tuning and prompt engineering both have an effective impact on high quality output of LLMs. In short, previous research showed that dependency on LLM's for annotation is being possible day by day. But still there are some barrier. For example LLM's are very sensitive over prompt and behave inconsistent in complex labeling. Depending on these problem our work proposed a fully automated annotation pipeline where prompt optimization, confidence-based filtering and comparison of model performance with human annotations across multiple NLP tasks. In a research, Nasution and Onan [21] has compared human annotation and LLM annotation in low resource languages like Turkish, Indonesian and Minangkabau. They do topic classification, sentiment analysis and emotion detection using ChatGPT-4, BERT and RoBERTa. From there work it is shown that in simple works, LLM's can perform good also better than human sometimes. But in complex case LLM's performance loses against human annotation. In recent years research [25] they make research on SLMs model to make more efficient and convenient. In this contrary they developed a pipeline using prompt engineering to reduce the cost. They gather total 150 leet code problem statement divided into 3 part which are easy, medium hard. To analysis this coding task they chosen four SLM models to outperform the result. Here this leetCode problem are solved by human. Then given to those models to make it efficient. So, they proposed four types of prompt engineering and they are zero shot, few shot, CoT and role prompting. Basically trying to figure out through prompt engineering can they get energy efficient problemsolving. Here each code is python code and human optimized and ask to them over those optimised code. They chose SLms which are smaller, designed to generate code and follow the instruction well which is very important (eg : Qwen2.5-coder-3B, Phi-3-Mini-4k etc). Here to benchmarking efficiency matrix they use runtime(ms), memory usage (KiB), Energy consumption(mWh) and all code was run 101 times per program. After that they figure out that some model cross human base line consistently and some are poor performing. From this they show Prompt engineering can reduce energy consumption, but not universally. SLMs react differently to the same prompts. Efficiency gains depend on model architecture and prompt type. Bigger or more complex prompting does not guarantee greener code

2.2.1 Related Research

In the recent past, Tomáš Horych et al [26] there they show classifier trained on an LLM-generated synthetic labels dataset can cope or be competitive with human labeled gold

standard dataset. They on a single task, which is media bias detection. By this task they also show llms can be low cost annotators without bias. Where our work is multiple model with multiple task. Zheng et al. [27] evaluated large language models (LLMs) as judges using GPT-4. They used two evaluation frameworks MT-Bench and Chatbot Arena, which are known for evaluating LLMs evaluation quality. They found some common GPT-4 biases such as position bias, verbosity, and reasoning limitations. Ultimately, they found that GPT-4 achieved over 80% match with human experts, which similar to human expert to human expert consistency. These findings indicate that LLM-based evaluation can be improved making them comparable to human annotators. Building on this, our work tries to apply the concept of LLMs as judges to LLMs as NLP annotators. In this approach, we evaluate how effectively LLMs can act as annotators by comparing several open-source models (with fewer than 10 billion parameters) across different tasks. We use two types of prompts: zero-shot and few-shot prompts.”Nemkova et al. [28], in their study, evaluated several state-of-the-art LLMs, such as GPT-3.5, GPT-4, LLaMA-3, Mistral-7B, and Claude-2, on binary classification tasks related to human rights violations in social media posts about Russia and Ukraine. They compared the LLMs with gold-standard annotated data and used zero-shot and few-shot prompts. They found that GPT-4 outperformed all other models. This work is very similar to ours, as we will also be using open-source LLMs and will employ zero-shot and few-shot prompts to evaluate around ten open-source models, such as DeepSeek and Phi, across eight different tasks..Akshay Goel et al. [29] the study is basicly focus on the single task which is medical information extraction from clinical notes. In this paper they collect those data and extract the data and lebelize them through large language llms model and then they recoorect the daat label using human check. Any halllucinated work llms removed. Then they train the downstream classifier train on the corrected dataset. So, on this piple line they used one task for labelizing and also used prompt engineering and ensembling.

2.3 Gap Analysis

These are gaps that we have identified from the previous research. The gaps are given below on the table:

Table 2.1: Gap analysis of our project on table 1

	Tomáš Horych et al [26]	Akshay Goel et al [29]	Our Focus
Goal	Detect & correct label errors	annotation with LLM + human	Evaluate LLMs as annotators
Task	1 tasks	1 task	8 type NLP tasks
Domain	General NLP	Clinical	Multi-domain NLP
Dataset	Noisy datasets	Clinical notes to lable	Human-labeled golden standard datasets
Models	Few LLMs	One LLM and human experts annotator	10 modern LLMs
Evaluation	Label error detection	annotation accuracy and cost	Annotation accuracy vs. human labels
Annotation	LLM verification	LLM assisted with human correction	Direct LLM annotation zero shot and few shot

Table 2.2: Gap analysis of our project on table 2

	Nemkova et al., [28]	Zheng et al., 2023 [30]	Our focus
Goal	Annotation evaluation on human-rights violation detection	Evaluating GPT-4 as judges	Evaluate LLMs as annotators across tasks
Task	binary classification	Conversational evaluation	8 structured NLP tasks
Domain	Social media	General text	Multi-domain NLP datasets
Dataset	Social media data	Crowdsourced data	Human-labeled benchmark datasets
Models	GPT-3.5, GPT-4, LLaMA, Mistral-7B, Claude	GPT-4	10 open-source models (<10B)
Evaluation	Annotation accuracy	Judgement accuracy	Annotation accuracy vs. human labels
Annotation	zero-shot and few-shot	not annotating	zero-shot and few-shot

2.4 Summary

We have reviewed around ten papers of them, we found four that are reasonably related to our project. However, even in those, we identified quite a few gaps. We are annotating in around eight tasks, and none of the papers cover more than one task with ten models. Secondly, our model choice is more diverse, as we are using ten open-source generative models. Additionally, our prompt choices are also diverse, as we employ zero-shot and few-shot. This makes our project more unique than all the other papers we have studied.

Chapter 3

Project Design

In this section we explain the project design of our research.

3.1 Detailed Methodology and Design

Our process starts with finding true human-labeled dataset to work on. After that we select tasks and models. Our next step is Prompt Configuration. We select 2 types of prompts. Then we extract labels from the output of the models. We use to finalize the generated label by the model. The overall workflow is illustrated below :

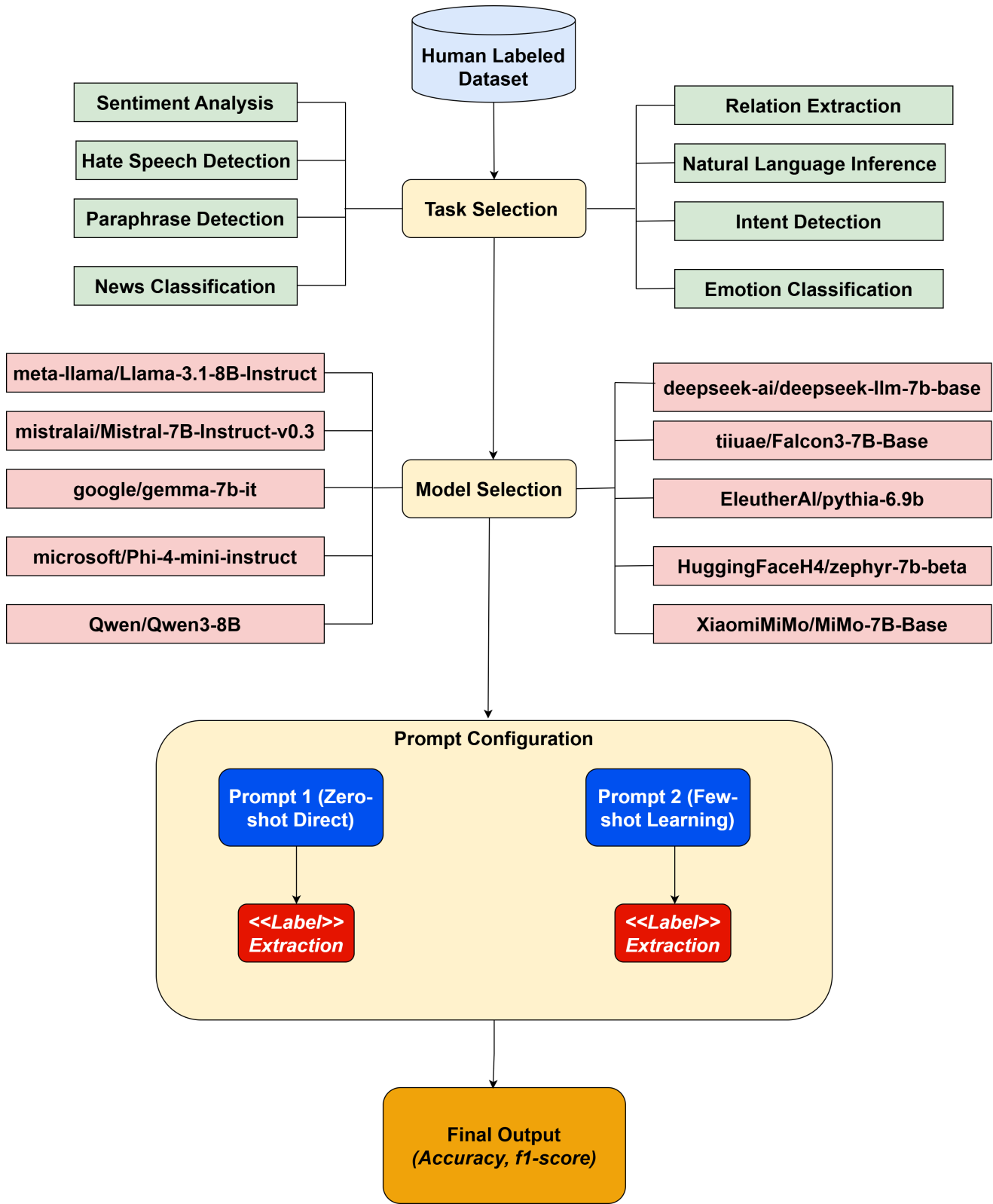


Figure 3.1: Workflow of our solution

3.1.1 Human Labeled Dataset Collection

One of the challenge of our project was to find true human annotated/labeled dataset. To do so we have read papers related to that particular dataset and ensured that the dataset are indeed human labeled. The datasets are snli, sem eval 2010 task 8, OLID, snips built in intents, PAWS, Ag news, IMDB, goemotions, conll2003.

3.1.2 Task Selection

Our next step was finding tasks that we can perform on those datasets. We have selected 8 tasks. Those tasks are Sentiment Analysis, Hate Speech Detection, Intent Detection, Paraphrase Detection, Emotion Classification etc.

3.1.3 Model Selection

After selecting dataset and task we select 10 LLM's. They are deepseek-ai/deepseek-llm-7b-base, meta-llama/Llama-3.1-8B-Instruct, mistralai/Mistral-7B-Instruct-v0.3, google/gemma-7b-it, microsoft/Phi-4-mini-instruct, Qwen/Qwen3-8B, tiuuai/Falcon3-7B-Base, EleutherAI/pythia-6.9b, HuggingFaceH4/zephyr-7b-beta, XiaomiMiMo/MiMo-7B-Base.

3.1.4 Prompt Ensemble Configuration

LLM models have limitations. They can hallucinate. Hallucination means when a model doesn't understand the prompt properly it will generate random output of it's own. This is a big problem because if the model could not get what we wanted from it then we might end up with false conclusion about the model. To stop this we basically used 2 types of prompting. These are :

Prompt 1 (Zero-shot Direct) : In zero shot no examples are given. Only task label names with their definition will be given.

Prompt 2 (Few-shot Learning) : We provide example instances for definition as well. By understanding the examples, models will be able to learn which instances actually belongs from which class.

Table 3.1: Zero-shot prompt templates used for each task.

	Sentiment Analysis	Hate Speech detection	Intent detection
Task instruction	You are a sentiment analysis expert. What is the sentiment of movie review above. Choose only one of the following: - Positive - Negative Respond with one word only:	Read this tweet below : Now tell me if the tweet is : - Offensive - Non-offensive Answer in one word only.	Determine the user's intent based on the sentence below. Select the most appropriate intent from the list: - ComparePlaces - RequestRide - GetWeather - SearchPlace - GetPlaceDetails - ShareCurrentLocation - GetTrafficInformation - BookRestaurant - GetDirections - ShareETA Answer in one word only
Target input	"The service was extremely slow and disappointing."	"All members of that community are criminals."	"Book a ride for me to the train station."
LLM's output	Negative	Hate	RequestRide
LLM's output	<label> Explanation: <explanation>	<label> Explanation: <explanation>	<label> Explanation: <explanation>

Table 3.2: Zero-shot prompt templates used for each task.

	Paraphrase Detection	Emotion Classification	News Classification
Task instruction	You are an expert in identifying whether two sentences have the same meaning or not. Classify the sentence pairs as: - 1 → Paraphrase (they express the same meaning) - 0 → Not Paraphrase (they express different meanings) Consider overall semantic meaning, not exact word matching. Sentence 1: "sentence1[0]" Sentence 2: "sentence2[0]" Answer with only 0 or 1. Answer:"""	You are an AI assistant trained to understand human emotions in text. Classify the emotion expressed in the sentence above as one of the following: - Anger - Joy - Sadness Answer in one word only.	You are an AI assistant trained to classify news articles based on their content. Classify the article into one of the following categories: - World - Sports - Business - SciTech Answer in one word only.
Target input	Sentence 1: "She bought a new laptop." Sentence 2: "She purchased a brand new computer."	"I can't stop smiling today, everything feels perfect!"	"The central bank announced new economic reforms to stabilize inflation."
LLM's output	1	Joy	Business

Table 3.3: Zero-shot prompt templates used for each task.

	Natural Language Inference	Relation Extraction
Task instruction	You are an expert in Natural Language Inference (NLI). Classify the relationship between the following pair as: - entailment - neutral - contradiction	Identify the single most accurate semantic relation between two entities from the following categories: 1. Cause-Effect: One causes the other. 2. Component-Whole: Part-whole relation. 3. Content-Container: One is contained in the other. 4. Entity-Destination: Entity moves to destination. 5. Entity-Origin: One comes from the other. 6. Instrument-Agency: Agent uses instrument. 7. Member-Collection: Belongs to a group. 8. Message-Topic: Communication about topic. 9. Product-Producer: Producer generates product. 10. Other: None of the above. Output only the relation label.
Target input	Premise: "Children are playing in the playground." Hypothesis: "Kids are outdoors."	"The doctor treated the patient using a new medicine."
LLM's output	entailment	Product-Producer

Table 3.4: Few-shot prompt templates used for each task.

	Sentiment Analysis	Hate Speech Detection
Task instruction	You are a sentiment analysis expert. Classify the sentiment of movie reviews. Review: "This movie was amazing, the acting was fantastic." Answer: Positive Review: "The plot was boring and I wasted my time." Answer: Negative Review: "Great visuals and excellent story." Answer: Positive Now classify the following review. Review: "{review[0]}" Choose only one: - Positive - Negative Respond with one word only. Answer:	You are an expert in detecting offensive language in tweets. Example 1: Tweet: "@user Shut up you stupid person." Label: Offensive Example 2: Tweet: "@user Thanks for sharing this information!" Label: Non-offensive Example 3: Tweet: "This is the worst thing ever, you are disgusting." Label: Offensive Example 4: Tweet: "Happy birthday! Wish you success." Label: Non-offensive Now classify this tweet: Tweet: "{review[0]}" Answer only: Offensive or Non-offensive Answer:
Target input	"The customer support was absolutely useless and disappointing."	"All people from that community are lazy and should not be trusted."
LLM's output	Negative	Hate

Table 3.5: Few-shot prompt templates used for each task.

	Intent Classification	Paraphrase Detection
Task instruction	Determine the user’s intent based on the sentence below. Select the most appropriate intent from the list: - ComparePlaces: Comparing two or more locations. - RequestRide: Asking to book or request a ride/transport. - GetWeather: Asking about weather conditions or forecast. - SearchPlace: Searching for a place or type of place. - GetPlaceDetails: Requesting details about a specific place. - ShareCurrentLocation: Sharing the user’s current location. - GetTrafficInformation: Asking about traffic conditions. - BookRestaurant: Making a restaurant reservation. - GetDirections: Asking for directions to a destination. - ShareETA: Sharing estimated arrival time. Answer in one word only. Examples: Sentence: "Which is better for families, Paris or Rome?" Answer: ComparePlaces Sentence: "Book me a ride to the airport." Answer: RequestRide Sentence: "What’s the weather like in Dhaka tomorrow?" Answer: GetWeather... Now classify the following sentence: Sentence: "{text[0]}" Answer:	You are an expert in identifying whether two sentences have the same meaning or not. Classify the sentence pairs as: - 1 → Paraphrase (they express the same meaning) - 0 → Not Paraphrase (they express different meanings) Here are some examples: Sentence 1: "She went to the grocery store." Sentence 2: "She visited the supermarket." Answer: 1 Sentence 1: "He enjoys playing guitar." Sentence 2: "He hates playing instruments." Answer: 0 Sentence 1: "The cat sat on the mat." Sentence 2: "A feline was sitting on the rug." Answer: 1 Now classify this pair: Sentence 1: "{sentence1[0]}" Sentence 2: "{sentence2[0]}" Answer:
Target input	"Can you book me a taxi to the airport?"	Sentence 1: "She bought a new car." Sentence 2: "She purchased a brand new vehicle."
LLM’s output	RequestRide	1

Table 3.6: Few-shot prompt templates used for each task.

	Emotion Classification	News Classification
Task instruction	You are an AI assistant trained to understand human emotions in text. Classify the emotion expressed in each sentence as one of the following: - Anger - Joy - Sadness Answer in one word only. Examples: Sentence: "I can't believe you lied to me after everything I've done for you!" Emotion: Anger Sentence: "I just got promoted at work today and I feel amazing!" Emotion: Joy Sentence: "I miss my grandmother so much since she passed away." Emotion: Sadness Now classify the following sentence: Sentence: "{sentence[0]}" Emotion:	You are an AI assistant trained to classify news articles based on their content. Classify each article into one of the following categories: - World: International politics, global events, diplomacy, or conflicts between nations. - Sports: Athletic events, teams, matches, tournaments, or player performances. - Business: Financial markets, companies, economy, trade, or corporate developments. - SciTech: Scientific discoveries, technological innovations, research, or digital advancements. Answer in one word only. Examples: Article: "Leaders from several countries met today to discuss climate agreements and rising geopolitical tensions in Eastern Europe." Answer: World Article: "The national team secured a dramatic last-minute victory in the championship final after scoring twice in extra time." Answer: Sports Article: "The stock market rallied today as major tech companies reported stronger-than-expected quarterly earnings." Answer: Business Article: "Researchers have developed a new artificial intelligence chip that significantly improves energy efficiency in data centers." Answer: SciTech Now classify the following article: Article: "{article[0]}" Answer:
Target input	"I am so excited about starting my new job tomorrow!"	"Scientists have developed a new battery technology that charges in under five minutes."
LLM's output	Joy	SciTech

Table 3.7: Few-shot prompt templates used for each task.

	Natural Language Inference	Relation Extraction
Task instruction	You are an expert in Natural Language Inference (NLI). The relationship between a Premise and Hypothesis can be: - entailment $\rightarrow$ The hypothesis is definitely true based on the premise. - neutral $\rightarrow$ The hypothesis might be true, but the premise does not give enough information. - contradiction $\rightarrow$ The hypothesis is definitely false based on the premise. Here are some examples: Example 1: Premise: "A man is playing a guitar on stage." Hypothesis: "A person is performing music." Answer: entailment Example 2: Premise: "A woman is reading a book in the park." Hypothesis: "The woman is a teacher." Answer: neutral... Now classify the following pair: Premise: "{premise[0]}" Hypothesis: "{hypothesis[0]}" Answer only one word: entailment neutral contradiction Answer:	You are an expert in semantic relation extraction. Identify the single most accurate relation between two entities marked as $e1_i \dots i / e1_i$ and $e2_i \dots i / e2_i$. Categories: 1. Cause-Effect: One causes the other. 2. Component-Whole: Part-whole relation. 3. Content-Container: One is contained in the other. 4. Entity-Destination: Entity moves to destination. 5. Entity-Origin: One comes from the other. 6. Instrument-Agency: Agent uses instrument. 7. Member-Collection: Belongs to a group. 8. Message-Topic: Communication about topic. 9. Product-Producer: Producer generates product. 10. Other: None of the above. Sentence: "{sentence[0]}" Output only the relation label. Answer:
Target input	Premise: "A man is riding a bicycle in the park." Hypothesis: "A person is outdoors."	"The chef prepared the meal using a sharp knife."
LLM's output	entailment	Instrument-Agency

Chapter 4

Implementation and Results

4.1 Environment Setup

All of our experiments were run on a Google Colab environment which had been given GPU acceleration to speed up model evaluating time. The models we used were downloaded from the Hugging Face Model Hub using the transformers library.

We ran all of our experiments in a Google Colab environment. When we needed high memory runtime environments they were also available. We used an T4 GPU with 15GB of VRAM for GPU acceleration which proving in google colab.

We used the following software libraries and tools which are Python 3.x , PyTorch , Hugging Face Transformers ,Datasets library ,Scikit-learn for computing accuracy and macro f1 score metrics.

Pandas and NumPy.

Model Access

All of the Large Language Models (LLMs) that we tested were loaded directly from the Hugging Face Model Hub by loading their official checkpoints. Inference was done without any additional fine tuning using zero shot and few shot prompt based inference strategies.

Evaluation

Model performance was evaluated using Accuracy and Macro F1-score. These were computed using the scikit-learn package.

4.2 Testing and Evaluation

We have used Accuracy and F1-score as our evaluation matrices.

Accuracy: Accuracy measures the model's overall correctness. It is estimated by the

ratio of accurately predicted instances to the total predictions, defined as,

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.1)$$

F1-score: F1-score measures the performance of classification models. We can calculate F1-score from the recall and precision estimation as follows,

$$F1 - score = \frac{2 * precision * recall}{precision + recall} \quad (4.2)$$

4.3 Results and Discussion

Here we are going to showing all the task basis model results base on a few-shot and zero-shot evaluation of the performance of each model.

Table 4.1: Zero-Shot vs Few-Shot Performance Comparison for Sentiment Analysis

Model	Zero-Shot		Few-Shot	
	Accuracy	F1	Accuracy	F1
Qwen/Qwen3-8B	0.9440	0.9421	0.9540	0.9520
meta-llama/Llama-3.1-8B-Instruct	0.9400	0.9380	0.9420	0.9390
tiiuae/Falcon3-7B-Base	0.9400	0.9375	0.9460	0.9440
microsoft/Phi-4-mini-instruct	0.9380	0.9350	0.9280	0.9230
google/gemma-7b-it	0.9320	0.9298	0.8840	0.8680
mistralai/Mistral-7B-Instruct-v0.3	0.9240	0.9185	0.8860	0.8700
HuggingFaceH4/zephyr-7b-beta	0.9020	0.8932	0.8500	0.8220
deepseek-ai/deepseek-llm-7b-base	0.8540	0.8298	0.8260	0.8490
XiaomiMiMo/MiMo-7B-Base	0.6280	0.3960	0.8000	0.7470
EleutherAI/pythia-6.9b	0.5080	0.6740	0.5080	0.0080

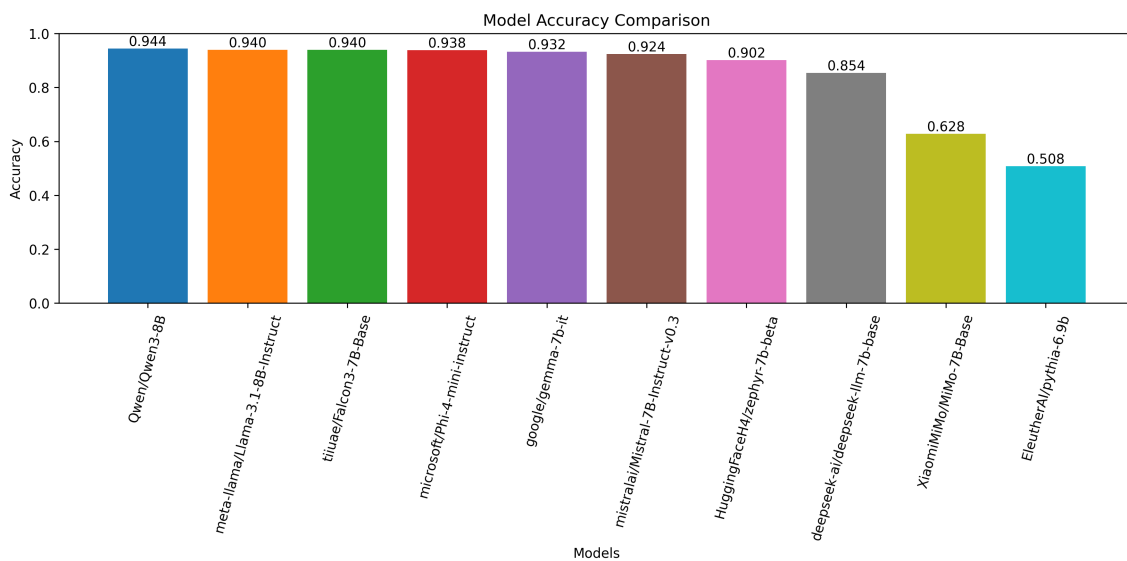


Figure 4.1: Bar-chart of accuracy using zero-shot prompt for Sentiment Detection

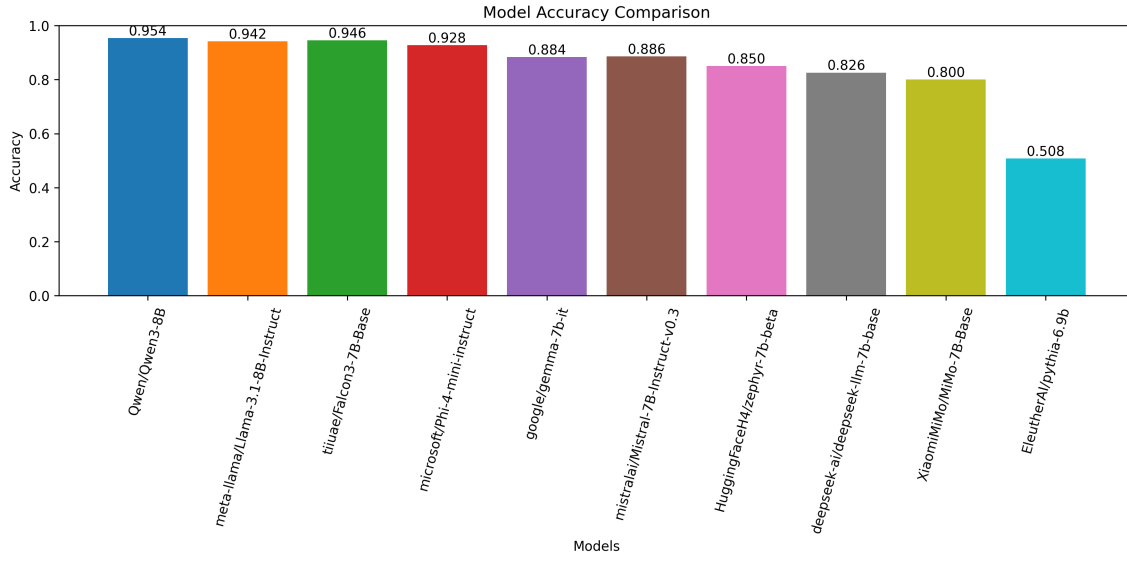


Figure 4.2: Bar-chart of accuracy using few-shot prompt for Sentiment Detection

Table 4.2: Zero-Shot vs Few-Shot Performance Comparison for Hate Speech Detection

Model	Zero-Shot		Few-Shot	
	Accuracy	F1	Accuracy	F1
Qwen/Qwen3-8B	0.7725	0.7786	0.7770	0.7910
google/gemma-7b-it	0.7525	0.7834	0.6225	0.7250
meta-llama/Llama-3.1-8B-Instruct	0.7525	0.7765	0.7400	0.7581
microsoft/Phi-4-mini-instruct	0.7200	0.6800	0.7875	0.7837
tiuae/Falcon3-7B-Base	0.7200	0.7295	0.7050	0.7590
HuggingFaceH4/zephyr-7b-beta	0.7125	0.6796	0.7100	0.6400
mistralai/Mistral-7B-Instruct-v0.3	0.6975	0.6739	0.7320	0.6930
XiaomiMiMo/MiMo-7B-Base	0.6925	0.7100	0.7075	0.7640
deepseek-ai/deepseek-llm-7b-base	0.6375	0.5748	0.4900	0.0377
EleutherAI/pythia-6.9b	0.5000	0.4565	0.5050	0.6510

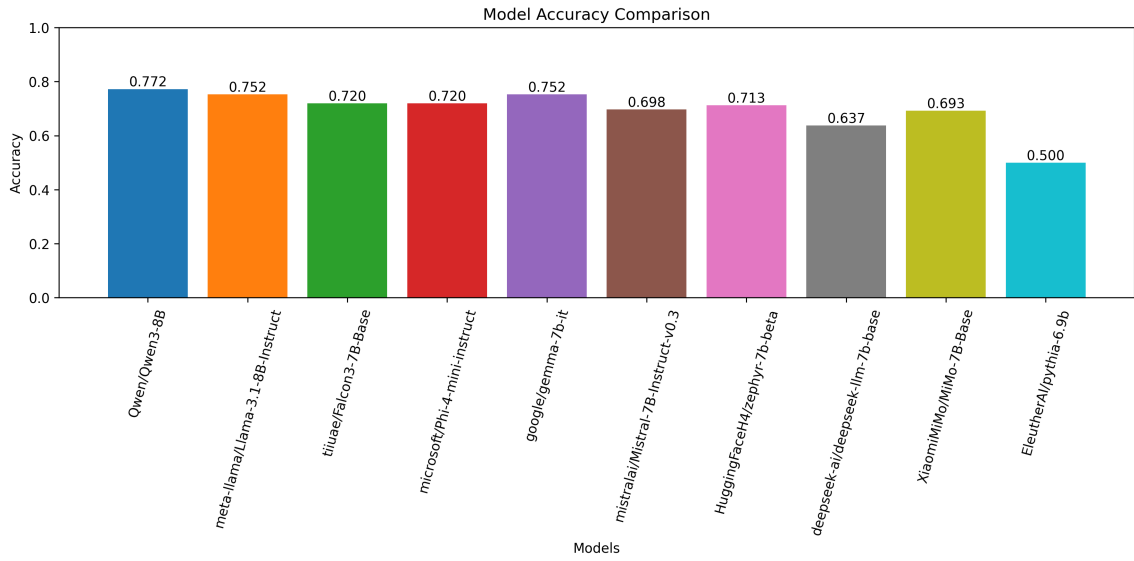


Figure 4.3: Bar-chart of accuracy using zero-shot prompt for Hate Speech Detection

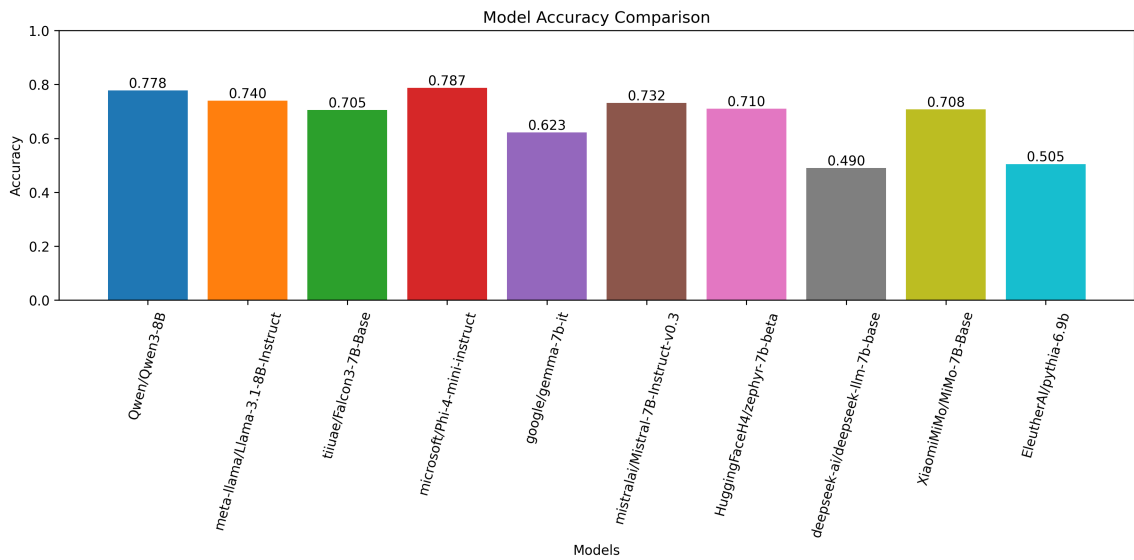


Figure 4.4: Bar-chart of accuracy using few-shot prompt for Hate Speech Detection

Table 4.3: Zero-Shot vs Few-Shot Performance Comparison for Intent Detection

Model	Zero-Shot		Few-Shot	
	Accuracy	F1-macro	Accuracy	F1
XiaomiMiMo/MiMo-7B-Base	0.9024	0.9052	0.0914	0.0719
Qwen/Qwen3-8B	0.8994	0.9031	0.3932	0.4010
tiiaue/Falcon3-7B-Base	0.8293	0.8511	0.9115	0.9161
HuggingFaceH4/zephyr-7b-beta	0.8109	0.8205	0.9146	0.9097
microsoft/Phi-4-mini-instruct	0.8049	0.8114	0.9512	0.9558
google/gemma-7b-it	0.7866	0.7852	0.2012	0.2030
mistralai/Mistral-7B-Instruct-v0.3	0.7805	0.7783	0.96036	0.9633
meta-llama/Llama-3.1-8B-Instruct	0.7713	0.7108	0.6310	0.6779
deepseek-ai/deepseek-llm-7b-base	0.7043	0.6299	0.7713	0.7900
EleutherAI/pythia-6.9b	0.6097	0.0209	0.3109	0.2762

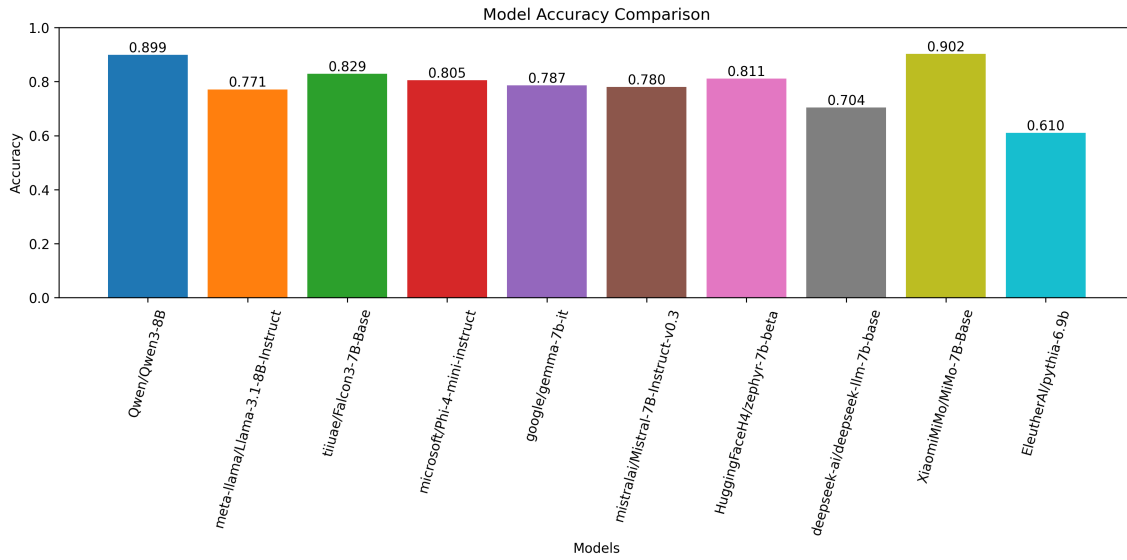


Figure 4.5: Bar-chart of accuracy using zero-shot prompt for Intent Detection

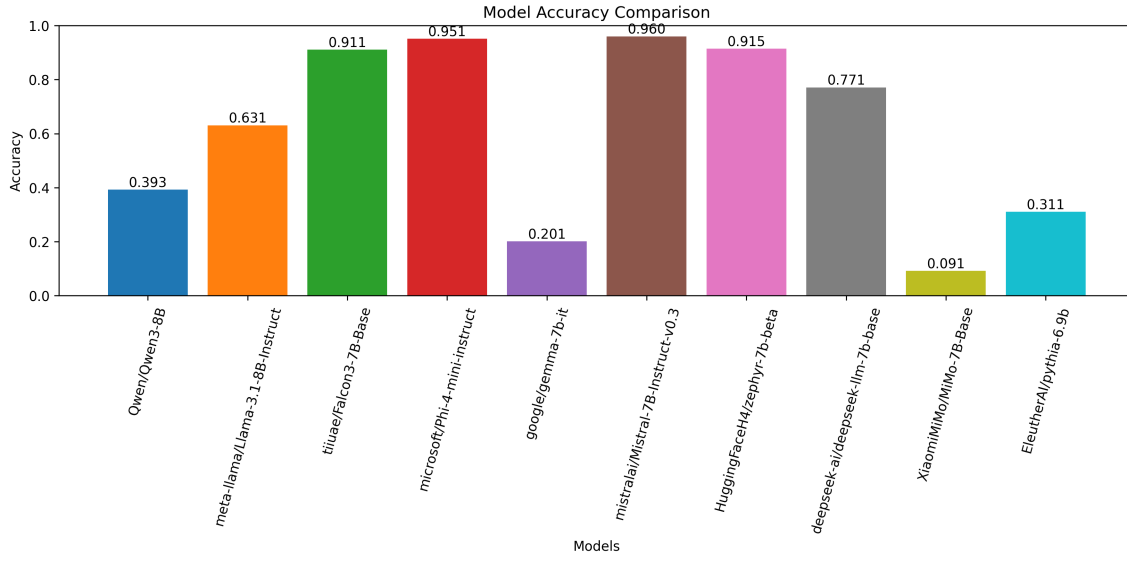


Figure 4.6: Bar-chart of accuracy using few-shot prompt for Intent Detection

Table 4.4: Zero-Shot vs Few-Shot Performance Comparison for Paraphrase Detection

Model	Zero-Shot		Few-Shot	
	Accuracy	F1	Accuracy	F1
tiuae/Falcon3-7B-Base	0.7800	0.8095	0.7500	0.7872
XiaomiMiMo/MiMo-7B-Base	0.7650	0.7982	0.7125	0.7619
HuggingFaceH4/zephyr-7b-beta	0.5475	0.6830	0.6625	0.7428
microsoft/Phi-4-mini-instruct	0.6925	0.7602	0.6625	0.7429
Qwen/Qwen3-8B	0.7075	0.7710	0.6275	0.6940
meta-llama/Llama-3.1-8B-Instruct	0.6675	0.7417	0.6225	0.7068
deepseek-ai/deepseek-llm-7b-base	0.5000	0.6660	0.5500	0.6886
mistralai/Mistral-7B-Instruct-v0.3	0.5200	0.6756	0.5350	0.6804
google/gemma-7b-it	0.5000	0.6660	0.5200	0.6571
EleutherAI/pythia-6.9b	0.5000	0.3300	0.5175	0.6485

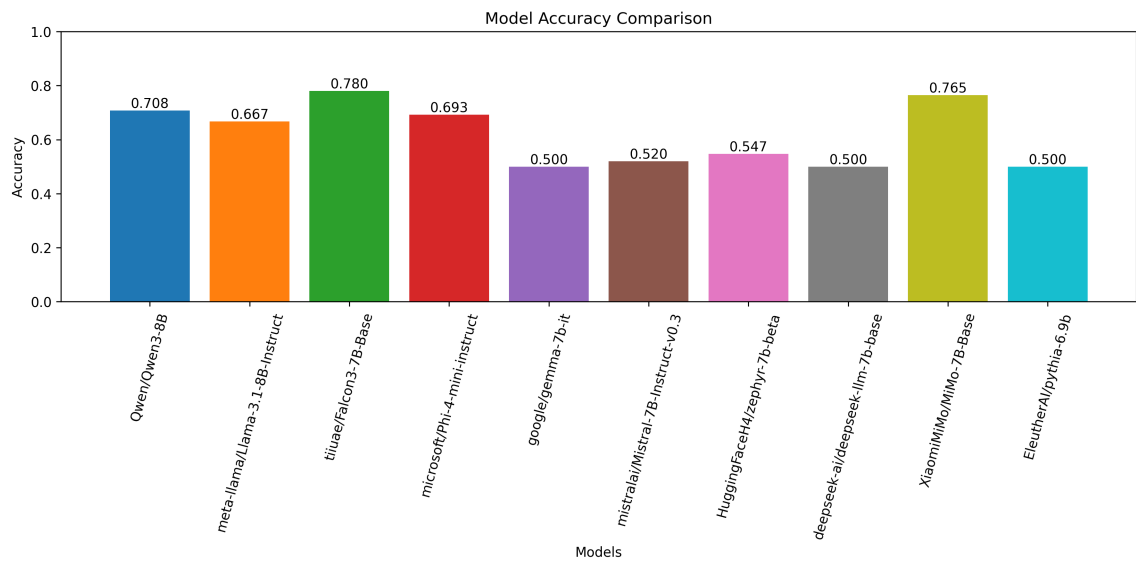


Figure 4.7: Bar-chart of accuracy using zero-shot prompt for Paraphrase Detection

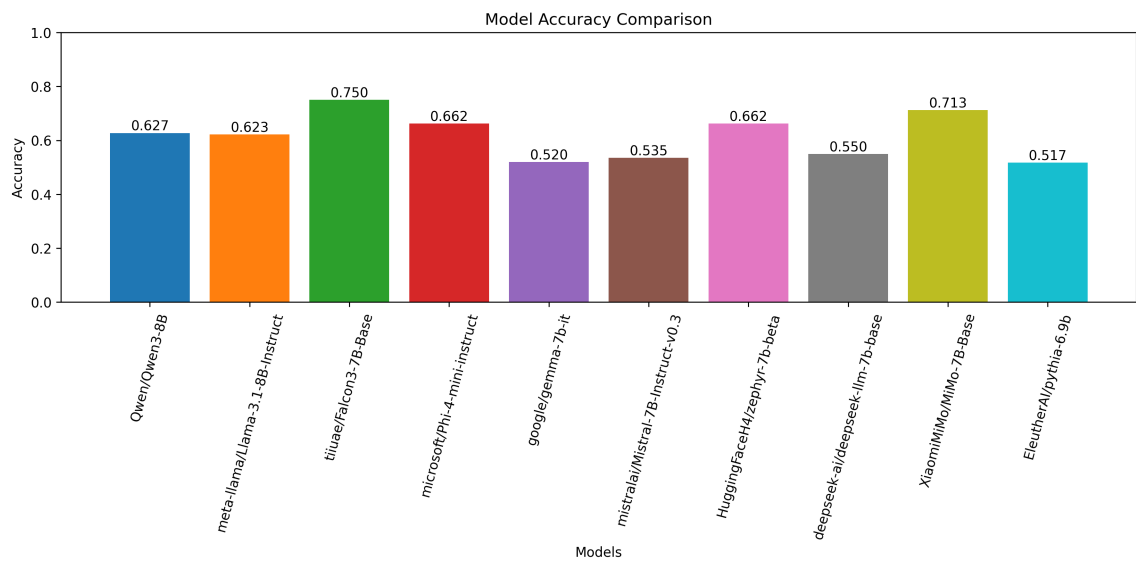


Figure 4.8: Bar-chart of accuracy using few-shot prompt for Paraphrase Detection

Table 4.5: Zero-Shot vs Few-Shot Performance Comparison for Emotion Classification

Model	Zero-Shot		Few-Shot	
	Accuracy	F1	Accuracy	F1
tiiuae/Falcon3-7B-Base	0.8383	0.8371	0.7516	0.7464
microsoft/Phi-4-mini-instruct	0.8383	0.8388	0.8266	0.8261
HuggingFaceH4/zephyr-7b-beta	0.8366	0.8371	0.7350	0.7310
mistralai/Mistral-7B-Instruct-v0.3	0.8300	0.8308	0.7866	0.7875
google/gemma-7b-it	0.8267	0.8274	0.5816	0.5518
meta-llama/Llama-3.1-8B-Instruct	0.8100	0.8107	0.7416	0.7415
Qwen/Qwen3-8B	0.7900	0.7932	0.4316	0.3399
deepseek-ai/deepseek-llm-7b-base	0.7867	0.7805	0.5680	0.5463
EleutherAI/pythia-6.9b	0.4250	0.3686	0.4333	0.3561
XiaomiMiMo/MiMo-7B-Base	0.3550	0.2113	0.4067	0.3043

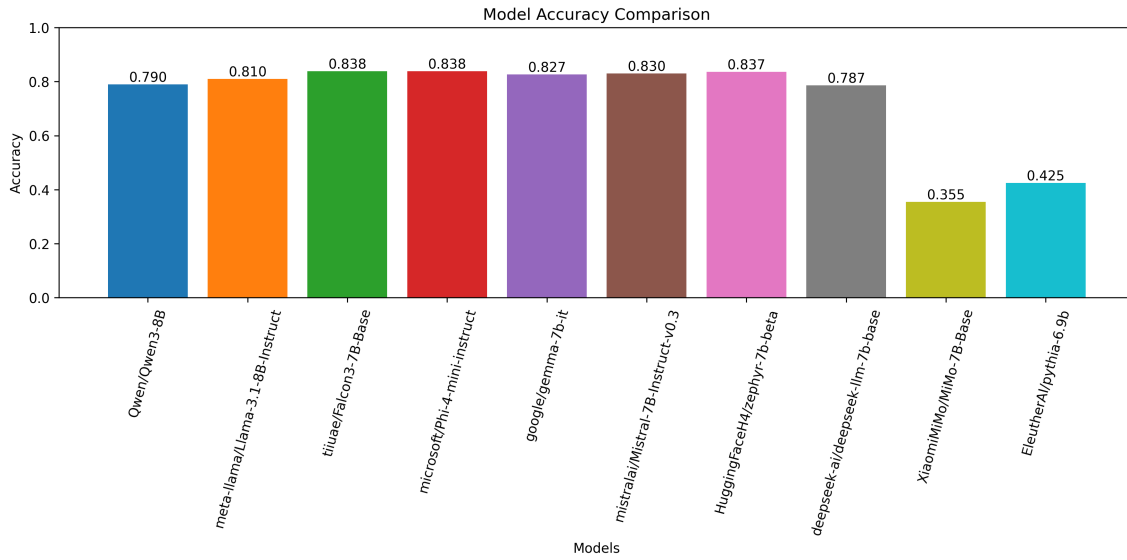


Figure 4.9: Bar-chart of accuracy using zero-shot prompt for Emotion Classification

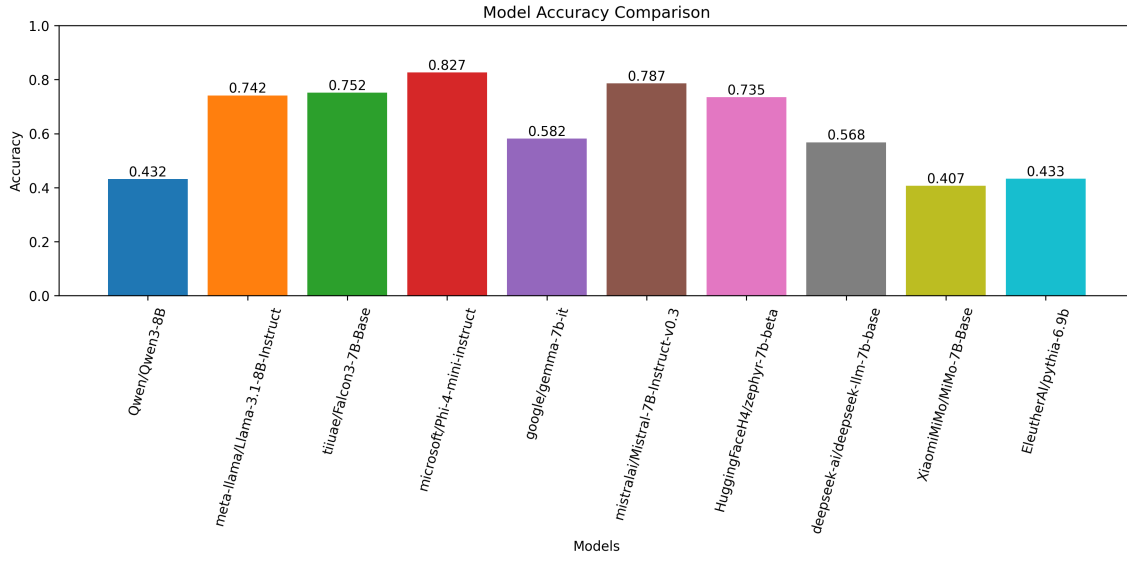


Figure 4.10: Bar-chart of accuracy using few-shot prompt for Emotion Classification

Table 4.6: Zero-Shot vs Few-Shot Performance Comparison for News Classification

Model	Zero-Shot		Few-Shot	
	Accuracy	F1	Accuracy	F1
Qwen/Qwen3-8B	0.8383	0.8366	0.7366	0.7340
XiaomiMiMo/MiMo-7B-Base	0.8050	0.7940	0.3966	0.3555
tiuae/Falcon3-7B-Base	0.7983	0.7899	0.7467	0.7270
meta-llama/Llama-3.1-8B-Instruct	0.7933	0.7848	0.7666	0.7532
google/gemma-7b-it	0.7717	0.7553	0.4633	0.4177
mistralai/Mistral-7B-Instruct-v0.3	0.7717	0.7553	0.7950	0.7874
microsoft/Phi-4-mini-instruct	0.7567	0.7415	0.7533	0.7415
HuggingFaceH4/zephyr-7b-beta	0.7566	0.7414	0.7266	0.7250
EleutherAI/pythia-6.9b	0.3750	0.3327	0.2516	0.1034
deepseek-ai/deepseek-llm-7b-base	0.3083	0.2403	0.5233	0.5114

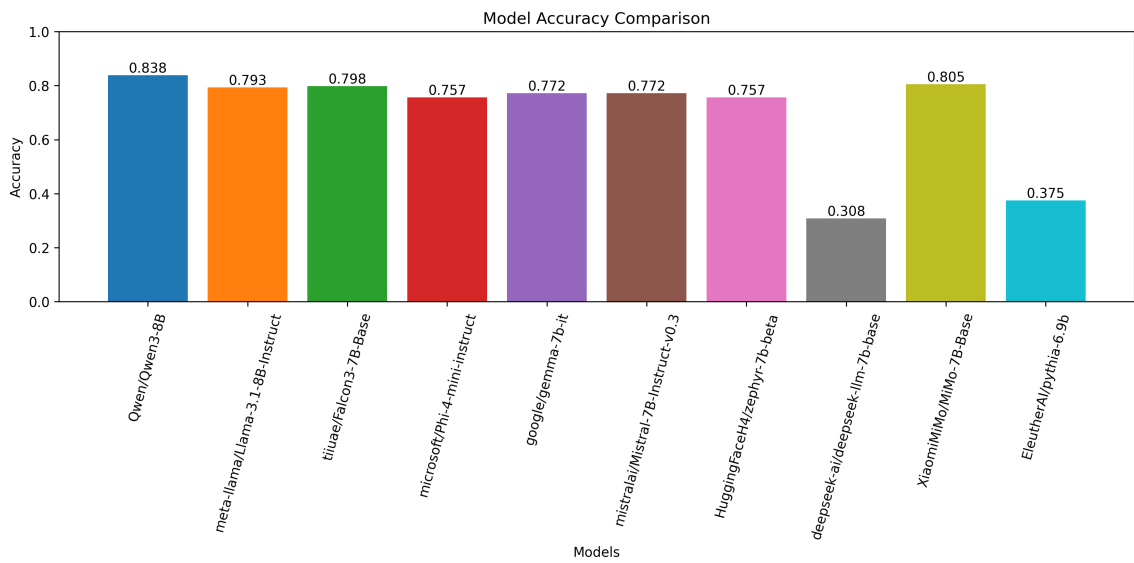


Figure 4.11: Bar-chart of accuracy using zero-shot prompt for News Classification

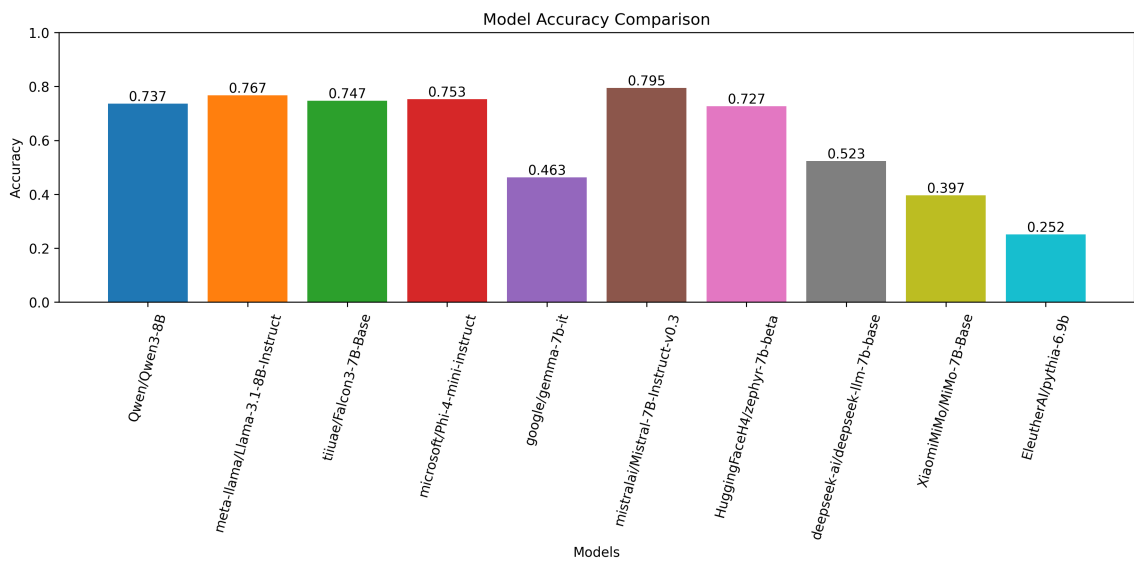


Figure 4.12: Bar-chart of accuracy using few-shot prompt for News Classification

Table 4.7: Zero-Shot vs Few-Shot Performance Comparison for Natural Language Inference (NLI)

Model	Zero-Shot		Few-Shot	
	Accuracy	F1 (Macro)	Accuracy	F1 (Macro)
microsoft/Phi-4-mini-instruct	0.8117	0.7992	0.8766	0.8772
HuggingFaceH4/zephyr-7b-beta	0.8116	0.8116	0.3316	0.1660
mistralai/Mistral-7B-Instruct-v0.3	0.6617	0.6544	0.5083	0.4592
Qwen/Qwen3-8B	0.6283	0.6307	0.6850	0.6540
XiaomiMiMo/MiMo-7B-Base	0.5966	0.7991	0.5133	0.4691
google/gemma-7b-it	0.5950	0.5547	0.5066	0.4541
meta-llama/Llama-3.1-8B-Instruct	0.4883	0.4240	0.4233	0.3458
tiiuae/Falcon3-7B-Base	0.4033	0.3032	0.4883	0.3963
deepseek-ai/deepseek-llm-7b-base	0.4033	0.3032	0.3333	0.1666
EleutherAI/pythia-6.9b	0.3166	0.2623	0.3333	0.1695

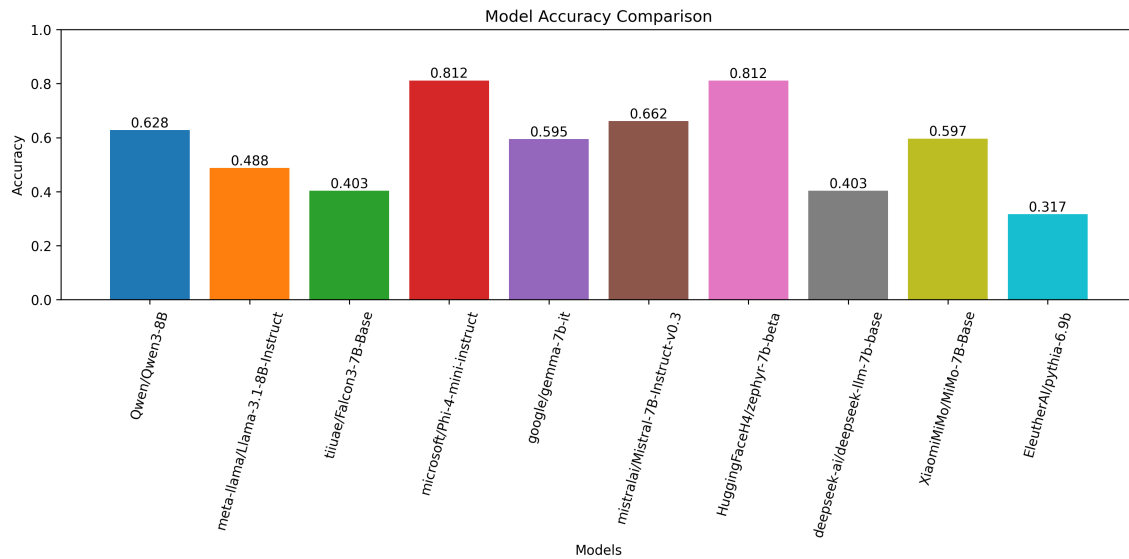


Figure 4.13: Bar-chart of accuracy using zero-shot prompt for Natural Language Inference

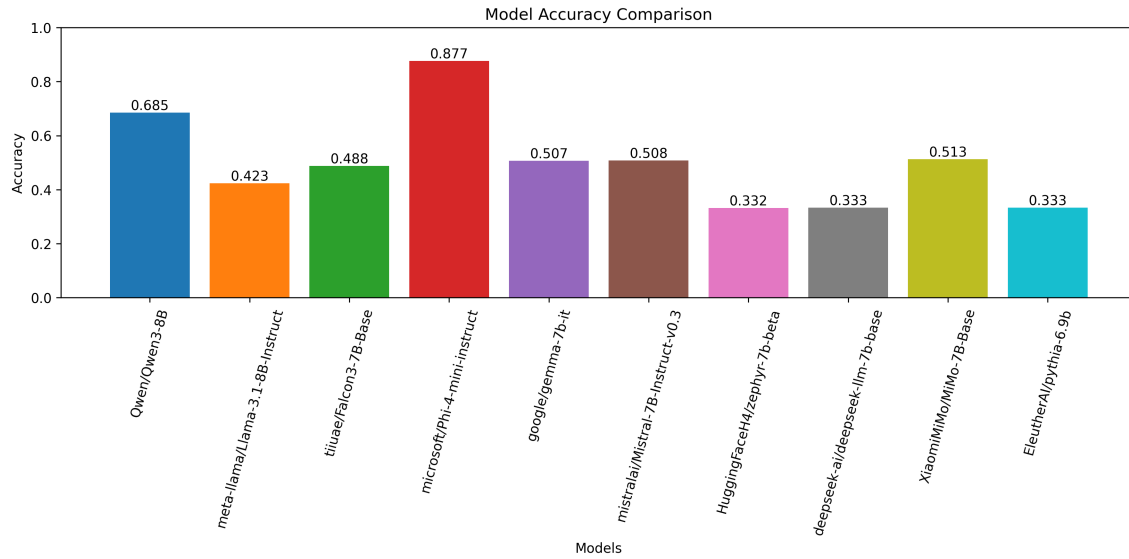


Figure 4.14: Bar-chart of accuracy using few-shot prompt for Natural Language Inference

Table 4.8: Zero-Shot vs Few-Shot Performance Comparison for Relation Extraction

Model	Zero-Shot		Few-Shot	
	Accuracy	F1 (Macro)	Accuracy	F1 (Macro)
google/gemma-7b-it	0.4080	0.3830	0.4740	0.4736
mistralai/Mistral-7B-Instruct-v0.3	0.4660	0.4674	0.4740	0.4705
microsoft/Phi-4-mini-instruct	0.3920	0.3776	0.4400	0.4239
Qwen/Qwen3-8B	0.4030	0.3957	0.4470	0.4385
HuggingFaceH4/zephyr-7b-beta	0.3430	0.3400	0.4300	0.4119
XiaomiMiMo/MiMo-7B-Base	0.3110	0.2990	0.3350	0.2992
meta-llama/Llama-3.1-8B-Instruct	0.2430	0.1931	0.3140	0.2992
tiuae/Falcon3-7B-Base	0.3110	0.3243	0.3100	0.3259
EleutherAI/pythia-6.9b	0.1710	0.0522	0.1770	0.0555
deepseek-ai/deepseek-llm-7b-base	0.1610	0.0347	0.1710	0.0568

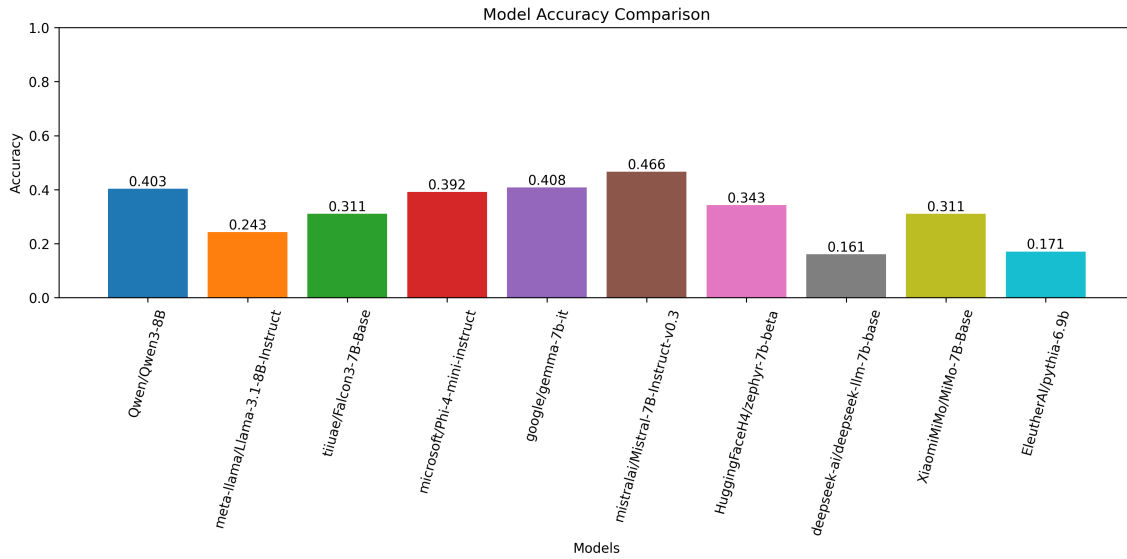


Figure 4.15: Bar-chart of accuracy using zero-shot prompt for Relation Extraction

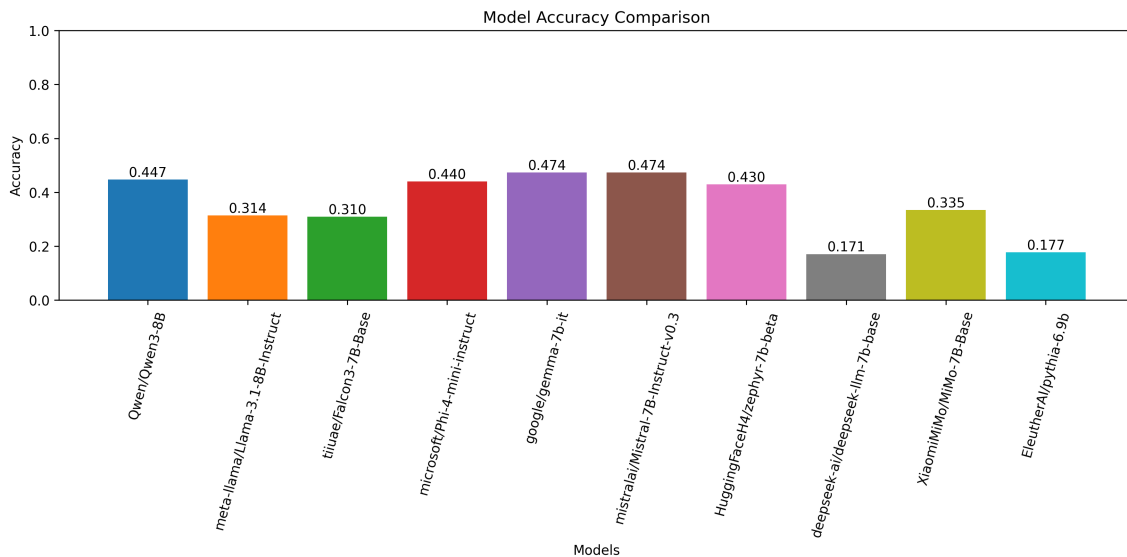


Figure 4.16: Bar-chart of accuracy using few-shot prompt for Relation Extraction

4.4 Summary

From the results we can see different LLMs perform differently in different tasks. Qwen/Qwen3-8B performed well in several tasks. According to the results of the experiments in sentiment analysis 4.1, it was noted that the performance of models varies across the range of zero-shot versus few-shot testing. The best overall performance award was given to Qwen/Qwen3-8B for both zero-shot and few-shot testing, however, there were still strong, stable performance results found from the Falcon3-7B and LLaMA-3.1-8B-Instruct models. In some cases, use of few-shot prompting improved the results of certain models.

Most hate speech detection models do well in table 4.2 with few-shot prompting based on their results. Both Qwen3-8B as well as Phi - 4 mini - instruct have shown significant performance increases with few-shot prompts. On the other hand, some models had decreased performance indicating prompt design can affect a model's performance. Here by performing few shot best result outcome in phi-4-mini which is 0.7875 accuracy in zero shot was qwen3-8B. Here in table 4.3, Intent Detection in zero shot prompt XiaomiMiMo /MiMo-7B-Base was best model which is 0.9024 and second best was Qwen/Qwen3-8B. After few shot operation significantly decrease accuracy of XiaomiMiMo/MiMo-7B-Base which is 0.0914. On the other hand mistralai/Mistral-7B-Instruct-v0.3 in zero shot was 0.7805 accuracy and it turn over into 0.96036 accuracy by few shot operation and it becomes best result for this task. After that in table 4.4, Paraphrase Detection for zero shot accuracy for tiuae/Falcon3-7B-Base is 0.78 and f1 score is 0.8095 but in few shot it decrease the result. This same kind of senario happend in lots of models. Moreover some models result were increased like for few shot HuggingFaceH4/zephyr-7b-beta accuracy 0.5475, f1-score 0.6830 and after few shot it accuracy 0.6625 and f1 score 0.7428. go though the next classification table 4.5, Emotion Classification here best result is at both tiuae/Falcon3-7B-Base and microsoft/Phi-4-mini-instruct which is 0.8383 by zero shot. In few shot none of models increase at that point. Here in few shot best models is microsoft/Phi-4-mini-instruct is 0.7866. But it decreased the accuracy by few shot from zero shot accuracy. In table 4.6, News Classification best accuracy outcomes after zero shot rather than few shot which is 0.8383 by Qwen/Qwen3-8B. Second best models is XiaomiMiMo/MiMo-7B-Base .But after few shot operation none of them cross this result. In table 4.7, few shot operation for NLI task microsoft/Phi-4-mini-instruct performed best result where in zero shot it was less and which accuracy is 0.8766 and f1 score 0.8772. In zero shot HuggingFaceH4/zephyr-7b-beta accuracy is 0.8116 which is second highest but it drops the accuracy after few shot prompt. In table 4.8, shows that relation extraction is a more difficult task than the other tasks with regard to the overall performance of the various systems - the performance across the various models is generally lower than what is reported for the other tasks. In most cases (except for Falcon3-7B), few-shot prompting generally provides a performance increase for the majority of models, particularly for Gemma-7B, Phi-4-mini, Qwen-8B, and Zephyr-7B. Mistral-7B-Instruct provides stable and competitive results in zero-shot and few-shot settings. LLaMA-3.1-8B-Instruct showed significant improvement with few-shot prompting, while Falcon3-7B showed very little change. Models such as Pythia and DeepSeek, which performed poorly, showed slightly greater gains. Overall, the use of few-shot prompting has produced moderate improvements in relation extraction, but relations extraction will still be a challenge for most of the models in this study.

Chapter 5

Standards and Design Constraints

5.1 Compliance with the Standards

5.1.1 Hardware Standards

- **Platform:** We have used google colab for our work. Which is free for everyone. We got a GPU that was given to us for a hour or two.
- **GPU:**
 - **NVIDIA Tesla T4 GPU:** This free GPU was 15GB. Which was very helpful for our work as LLMs need GPU power for fast response. All the LLM model that we used have been inferenced by using this free GPU by google colab.
- **RAM:** Our google colab RAM was 12.7GB. This RAM was useful for downloading the LLM and store it. The models were downloaded from Hugging Face directly and stored in this RAM.
- **Storage:** The Hard Disk storage that we got was 112.6GB. This Hard Disk was also helpful for our work because whenever RAM couldn't handle much data it used Hard Disk. Also for saving results hard disk was useful.

5.2 Design Constraints

5.2.1 Ethical Constraint

While using the models and datasets we have taken permission from hugging face so that no license violations occur. All the models were used by ethical guidelines.

5.3 Cost Analysis

Category	Human Annotation (Students)		LLM-Based Annotation (Free Colab)	
Cost	No direct monetary cost (student labor unpaid)		No direct monetary cost (free-tier Colab + open-source models)	
Annotation Speed (team of 4)	240 samples/hour (assuming 60 seconds per sample)		1,000–2,000 samples/hour per model	
Time for 1,000 samples	4.2 hours		0.5–1 hour	
Time for 100,000 samples	420 hours		12–20 hours (parallelized among 4 members)	
Practical Limitations	Fatigue, slower speed, human errors		Colab runtime limit (~12 hours/session), RAM limits (12–16 GB)	

Figure 5.1: Cost Analysis

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

P1: Cannot be solved without deep engineering knowledge **Supports:** K3, K4, K5, K6, K8

- **K3 (Engineering Fundamentals):** We needed to understand how language models work. Also, we need knowledge about how those models make annotations using tokens and probabilities. It helps us to understand those models instead of only using them.
- **K4 (Specialist Knowledge):** In our project, we have worked on Sentiment Analysis, Natural Language Inference, Relation Extraction, etc. Each task has its own method or rules. Understanding them helps us to evaluate those models properly.
- **K5 (Design Knowledge):** We designed a system that integrates: 10 different models, 10 tasks, 3 prompt ensembles and Evaluation pipeline. This required architectural thinking and modular design.
- **K6 (Engineering Practice):** We have used Hugging Face, Python, and Google Colab for run those models and generate output. This was hands-on engineering work.
- **K8 (Research Literature):** We have studied more than 15 papers to design a scientifically correct benchmark. For this, we needed to study annotation quality, prompt engineering, and LLM behavior.

P2: Involves conflicting technical issues Supports: K5, K6

- **K5 (Design Knowledge):** We needed to balance different models, like some that are slow but good, and some that are speedy but accurate. We needed to create the best prompt and compare those models properly. This required thoughtful, constraint-aware design.
- **K6 (Engineering Practice):** First, we have to choose 15 LLM. After that, we face lots of problems like version issues, weight loading, prompt format, GPU Memory, etc. And finally, we selected 10 models for our project. All of these challenges we face that were needed to be solved using real-world engineering skills.

P3: Requires abstract thinking and modeling Supports: K2, K4

- **K2 (Mathematics & Statistics):** We calculated accuracy, precision, recall, and F1 across 8 tasks $\times$ 10 models = 80 comparisons. After that, using these parameters, we have to justify those models efficiency.
- **K4 (Specialist Knowledge):** All faults are not the same in NLP. Like wrong assumption in Name detection is one type of fault, and wrong assumption in sentiment is another type of fault. If we understand these issues, then we will be able to judge models properly.

P4: Involves rarely encountered issues Supports: K3, K4

- **K3 (Engineering Fundamentals):** LLMs can hallucinate. To stop the hallucination we have introduced Prompt Ensemble Configuration. This required deep thinking and problem solution which covers engineering fundamentals.

P5: Outside existing standards or codes Supports: K7

- **K7 (Ethics & Responsibility):** Our project creates an impact on researchers, students, and developers. Some want speed, and others care about accuracy. We balanced this by using free tools and covering many tasks.

Table 5.1: Mapping with complex problem solving.

P1 Dept of Knowl- edge	P2 Range of Con- flicting Require- ments	P3 Depth of Analysis	P4 Familiarity of Issues	P5 Extent of Applicable Codes	P6 Extent of Stake- holder Involve- ment	P7 Inter- dependence
✓	✓	✓	✓	✓		

5.4.2 Engineering Activities

Our work involves a large-scale experimental setup where open-source LLMs are employed to annotate data and are evaluated against human-annotated benchmarks using standard NLP metrics. This process involves planning, automation, statistical analysis, and reporting (Table 5.2).

Table 5.2: Mapping with complex engineering activities.

A1 Range of re- sources	A2 Level of Interac- tion	A3 Innovation	A4 Consequences for society and environment	A5 Familiarity
✓		✓		

A1: Range of Resources

The models that we have used had to be downloaded from Hugging face website. Also the datasets were downloaded from here as well. This satisfies range of resources.

A3: Innovation

To stop model unpredictability we taken some new ideas of our own. 2 types of prompt were introduces. They are Zero-shot Direct and Few-shot Learning.

5.5 Summary

This project clearly qualifies as a complex engineering problem beacuse It covers (P1), (P2), (P3), (P4), (P5). Each of these also includes the core knowledge profiles K1 to K8. It also satisfies engineering activities (A1) and (A3). So clearly our project solves a complex engineering problems.

Chapter 6

Conclusion

6.1 Summary

In our project, we have found out if LLMs size of 7B parameters or less can be used to annotate for common NLP tasks. For this, we have selected a total of 8 tasks and tested them with 10 SLMs.

Across all experiments:

- Some models are very good at easy tasks like sentiment analysis, intent detection, and news classification getting accuracy over 80 percent, reaching 90 percent for some tasks
- For complex tasks like "Natural Language Inference & Relation Extraction," the LLMs didn't perform well.
- Our research shows that LLMs can still annotate in many tasks in a few times with less effort. Improving on these models can make annotation easier and accurate.

Overall small size of LLM can perform as a human in many tasks and can be improved in the future for complex tasks also.

6.2 Limitation

Even though the results were good, some limitations were found :

1. Hardware Limitation: Since we have used Google Colab so we were not able to test large LLMs.
2. Dataset Size Reduction: Basically, we have used 500-1000 data so the result was slightly affected by statistical reliability. Using of full dataset can lead to a more accurate output.
3. Prompt Sensitivity: The Output result gets changed after a small change in the prompt.
4. Weak Performance on Complex Reasoning Tasks: LLMs cannot hit our expected result in complex tasks like NLI & RE.

5.No Fine-Tuning: Models were used in fully pretrained form. Fine-tuning might improve accuracy, but it was not feasible under our current limitations.

6.3 Future Work

Our project is in an initial phase for automated annotation using open-source LLMs right now. There are still many useful future research directions :

1)Utilization of Full Dataset: We want to make benchmark using the dataset. This will give us a more accurate view of a models performance.

2)Test on More Tasks: Testing SLMs performance to annotate on other NLP and even non-NLP tasks.

3)Use a Larger Model: Using bigger LLMs like 13B,30B, or even bigger LLMs can generate more accurate output.

4)Add Banglish / Low-Resource Languages: For Banglish, there is a possible way to create an annotation system.

5)Build a Web Application for Annotation: A user-friendly web app can allow users to upload datasets and automatically receive LLM-generated annotations.

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